Dear **VLA** Project Groups,

The goal of this project is to build a vision-language-conditioned manipulation policy that picks up an object and places it at a target location specified by a natural language prompt. You will be evaluated on three setups of increasing difficulty: direct color-conditioned pick-and-place, compositional instruction following, and placing an object on the image of a famous person.

Logistics

Join the course Slack. Specifically the channel named "project-1-vla". As well as the (closed) per-project-group channels "project-1-vla-group-XX". All further information and clarifications will only be shared on the common project-1 Slack channel. These channels are a space to ask open questions. The TAs will be able to answer your questions a lot faster this way and it will keep the Thursday session open for difficult debugging problems.

The following restrictions on the model architecture or training procedure are enforced:

- The architecture for each task (or combined) must be considered a VLA and use a pretrained vision language backbone. It does not need to contain a continuous action head or other extensions necessarily.
- The model must be learned and you must be able to show you trained (e.g. fine-tuning) on your own with the provided compute.
- You are free to add features such as RL post-training (not at demo day) or synthetic data generation to increase success rates if you want to.
- Training data can be sourced from publicly available resources or generated through teleoperation or synthetic means.
- Different models may be used across the three eval setups. But you can also use the same model to solve all tasks. (or the same model with different weights/checkpoints)
- You can unmount the provided SO101 camera from the wrist and use it as a agent/shoulder camera with a custom-built mount (which you can bring to the eval day) if you think this will improve your policy. But only a single camera can be used. For each eval (1,2,3) a different camera setting can be chosen.
- At intefernce/test time (at the demo day) you are not allowed to use any YOLO, image recognition, API calls to large VLMs in the cloud etc. for your policies. But you are allowed to use these models to create annotated training data for your VLM/VLA training before the demo day if you wish to do so.
- All policies should have a CLI interface where we (the TAs) can write the task instruction on demo day and then press "Enter" to run the policy rollout.

Please do not use any custom hardware except for what was provided in the box you received from the course. Treat the hardware with care, as it must be returned fully functional after completion of the course.

The manipulation objects will be provided by us.

- bowls (3 bowls per team, one of each color),
- squishy banana (one per team),
- coke can (one per team)

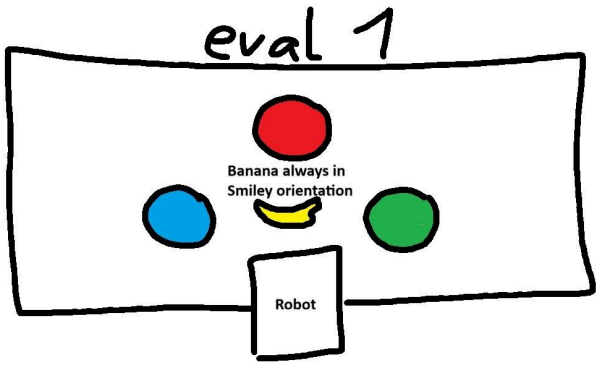
You will be able to pick them up early next week on your own or in the TA session on April 30th. For more up-to-date information join the Slack.

Evaluation

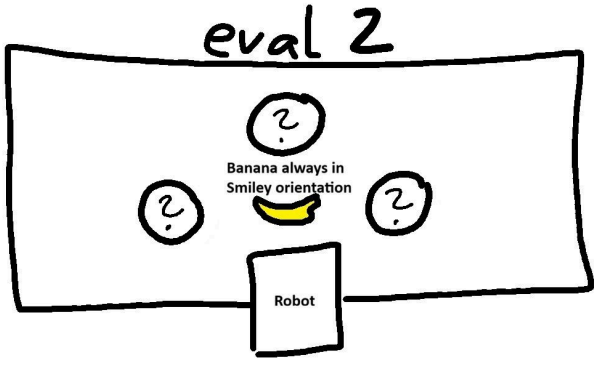
For the main evaluation, the environment will be standardized: the robot will be mounted on the edge of a white table, with objects placed in front of it as described below. You may choose exact object locations and reproduce them on eval day, but all objects must remain movable by approximately +-5cm from your chosen positions. The gripper of the robot must be at least ~15cm away from the banana at the beginning and the camera needs to have all three bowls and the banana in its field of view. The bowls are not allowed to be taped to the table. No mat/yoga mat or curtains or anything of this type is allowed to be added to the workspace.

The project goal is defined by the following milestones, with a maximum of 150/150 points available for the main evaluation (+50 points Bonus):

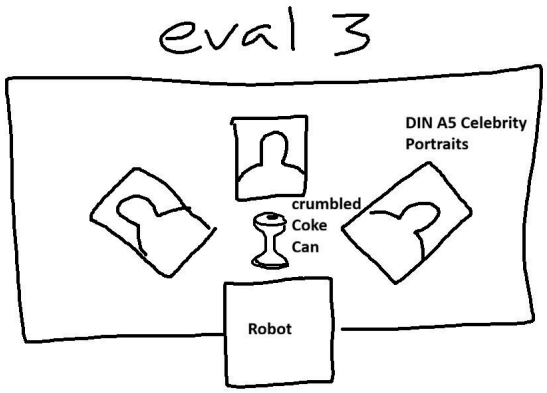
- **Eval 1 (50 pts):** Put the banana in the correct bowl. Three bowls will be placed in a semicircle in front of the arm, in the order blue, red, green from left to right. A small, squishy toy banana will be placed horizontally (smiley orientation) directly in front of the arm. Your policy will receive a prompt of the form "Put the banana in the [blue/red/green] colored bowl." Or a very similar prompt. Success is defined as the banana ending up inside the correct bowl within 20 seconds. We will perform 9 rollouts per group, three per color, with ~5.55 points per successful rollout.



- **Eval 2 (50 pts):** Follow compositional instructions with the same bowl setup as Eval 1 (but varying colors). Prompts will require reasoning beyond direct color lookup, e.g. "Put the banana into the 2nd bowl from the left from the robot perspective", "Put the banana into the bowl on the right of the red bowl from the robot perspective", "Put the banana into the bowl that is not green and not blue." (this would be the red bowl). The exact prompts will not be made known in advance but will be identical across groups and very similar to the ones above. Multiple different prompts will be evaluated, some easier, some harder. Points will be distributed evenly across rollouts. 20 seconds time per rollout to complete the task. The Banana will also be positioned in the middle of the bowls in a smiley orientation the same way as in Eval 1.



- **Eval 3 (50 pts):** Place an empty coke can on the correct celebrity image. DIN A5 color prints of celebrities will be placed in a semicircle in front of the robot, with a 330ml slim coke can standing in the middle. The coke bottle can be crumbled together on the sides, but still needs to be able to stand. You can take your exact coke bottle with you to the demo day. Your policy will receive a prompt referring to one of the people (Prompt: “Place the coke on [celebrity name]”), and must place the can on top of the correct image.
- You received (on Slack) 5 Taylor Swift Images, 5 Barak Obama images, 5 Yann LeCun images. These are your known in-distribution images or also called the "TOY" (Taylor Obama Yann) dataset. There will be 3 eval runs at the demo day with random combinations of 1 Swift, 1 Obama, 1 LeCun known in-distribution images in random order on the table. Print out the attached PDF with the images (in color), cut the images out so there is no more white border and these are the exact images we will use at eval day for these first 3 runs of eval 3.
- The second 3 runs of eval 3 will still be the same task, with Swift, Obama and LeCun but this time the images will be images of these celebrities from a held-out-in-distirbution dataset. E.g. we will use some Swift, Obama and LeCun images we haven't handed out to you.
- The last 3 runs will be out-of-distribution images of celebrities not known to you. (but very popular celebrities)
- The total 50 points for eval 3 are divided by 9 so each successful run gives you 5.55 points.
- 20 seconds time per rollout to complete the task.



- **Bonus:** Smallest model (competition between the teams). Model size is measured by total (active) parameter count of the model(s) used during inference for that eval.
    - Eval 1: **10/9/8/... pts** for 1st/2nd/3rd/... place of all VLA teams.
    - Eval 2: **20/18/16/... pts** for 1st/2nd/3rd/... place of all VLA teams.
    - Eval 3: **20/18/16/... pts** for 1st/2nd/3rd/... place of all VLA teams.

Recommended workflow to get started:

1. Calibrate robot setup.
2. Get teleoperation working.
3. Record and check 1-20 replay teleop episodes for correctness. (in LeRobot format)
4. Research small VLA architectures.
5. Implement basic VLA training loop and VLA model loading.
6. Setup compute access and try training any small model on any data.
7. Train on a single teleop episode and verify the model learns. (“overfit test”)
8. Collect teleop data for Eval 1 and train.

General Tips and Tricks:

- Be mindful of your compute, budget it wisely.
- Validate your teleop data before training.
- Model training can be slow, make sure to start debugging with a single episode early enough before moving to training on many episodes of your teleop data.
- Use LeRobot format to collect teleoperation data, using Huggingface's ecosystem buys you a lot of time.
- Reuse OSS code such as simple VLA training repositories and pretrained VLM/VLA models.
- Great starting points for your VLAs could be "FlowerVLA", "SmolVLA", "Smol-0-VLA", "TinyVLA"
- Gather your data in different lighting conditions, different tables etc. to create a more robust policy.
- The final evals will take place in HG, we suggest collecting at least some of your training data in this setup! (the tables are also not perfectly white)
- Think amount augmentations to your data to extend it.
- You may use cheaper GPUs than H100s to extend your allocated compute. For example, if you try to reach a batch size of 64 you could likely use 1xA100 with 80GB VRAM and gradient accumulation 2 to fit training onto a single GPU which reduces a lot of complexity and uses your GPU hours efficiently.
- Collect teleop data with your objects in slightly different positions (to generalize to the +-5cm they could move), different lighting levels and different backgrounds/tables.
- Many teams seem to be collecting and training with the full resolution images of your cameras. This will likely slow you down a lot. We recommend downsizing them to e.g. 256x256 or similar. Many powerful VLAs run at this camera resolution as it is very efficient and still performs just as well.
- Think about adding image augmentations such as random illumination changes to your policy training as this can help making your policy generalize better to unseen lighting situations (possible at demo day)

Best regards,
Project VLA TAs